Motion and graphs 1 1

Friday, September 12, 2025 9:33 AM

Kinematics: The study of motion without considering the forces that produce it.

Dynamics: The study of the causes of motion.

Scalar quantities: A quantity that has magnitude but not a direction.
↳ Ex: distance, speed

Vector quantities: A quantity that has both magnitude and direction.

position ($\vec{d}$): The straight line distance and direction of an object from a reference point.
→ Vector.

displacement ($\Delta\vec{d}$): Change in position.
→ Vector.

→ $\vec{\Delta d} = \vec{d}_2 - \vec{d}_1$

position 2 pos 1

Speed (v): How fast an object is moving at an instance
↳ Can only be positive or 0.
↳ The magnitude of the slope of the tangent to a Pos-time graph.

Avg speed (v_{av}): The total distance traveled divided by the total time to travel the distance.

$v_{av} = \frac{\Delta d}{\Delta t}$

Velocity ($\vec{v}$): The change in position ($\Delta\vec{d}$) AT THAT INSTANCE divided by the time interval (Δt)
→ The slope of the tangent on a pos-time graph

Avg velocity ($\vec{v}_{av}$): The displacement divided by the time interval for that change.
→ It is the slope of a Secant on a pos-time graph

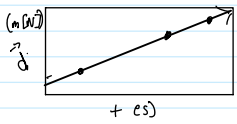
Time Interval: $\Delta t = t_2 - t_1$

Secant: A straight line connecting 2 seperate points on a curve

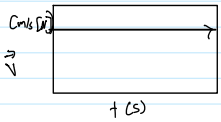
$* \quad \vec{v}_{av} = \frac{\Delta\vec{d}}{\Delta t} = \frac{d_2 - d_1}{t_2 - t_1}$

Graphs:

Position v Time



Velocity v Time



Tangent: A straight line that intersects a curve at a point and has the same slope as the curve at the point of intersection.

Average Acceleration ($\vec{a}_{av}$): The change in velocity divided by the time interval for that change

Instantaneous acceleration ($\vec{a}$): The acceleration at a particular instance in time.

$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t}$

* The slope of an object's velocity - time graph gives the object's acceleration

Practice Questions

Practice

1. A woman leaves her house to walk her dog. They stop a few times along a straight path. They walk a distance of 1.2 km [E] from their house in 24 min. In another 24 min, they turn around and take the same path home. Give your answers to the following questions in kilometres per hour. T/I A
   (a) Determine the average speed of the woman and her dog for the entire route. [ans: 3.0 km/h]
   (b) Calculate the average velocity from their house to the farthest position from the house. [ans: 3.0 km/h [E]]
   (c) Calculate the average velocity for the entire route. [ans: 0.0 km/h]
   (d) Are your answers for (b) and (c) different? Explain why or why not.
2. A bus driver begins a descent down a steep hill and suddenly sees a deer about to cross the road. He applies the brakes. During the bus driver's 0.32 s reaction time, the bus maintains a constant velocity of 27 m/s [forward]. Determine the displacement of the bus during the time the driver takes to react. T/I [ans: 8.6 m [forward]]
3. Drivers at the Daytona 500 Speedway in Florida must complete 200 laps of a track that is 4.02 km long. Calculate the average speed, in kilometres per hour, of a driver who completes 200 laps in 6.69 h. T/I [ans: 1.20×10^2 km/h]
4. A student in a mall walks 140 m [E] in 55 s to go to his favourite store. The store is not open yet, so he walks 45 m [W] in 21 s to go to another store. Calculate his
   (a) average speed [ans: 2.4 m/s]
   (b) average velocity T/I [ans: 1.3 m/s [E]]
5. A delivery truck heads directly south for 62 km, stopping for an insignificant amount of time, and then travels 78 km directly north. The average speed for the entire trip is 55 km/h. K/U T/I C
   (a) Determine the average velocity for the entire trip in kilometres per hour. [ans: 6.3 km/h [N]]
   (b) Why is the average velocity of the truck so much smaller than the average speed?

P.

1)

Practice

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   (c) Calculate the average velocity for the entire route. [ans: 0.0 km/h]
   (d) Are your answers for (b) and (c) different? Explain why or why not.

1) $\xrightarrow{1.2\ km\ [E]}$ $t = 24\ min$

$\xleftarrow{1.2\ km\ [W]}$ $t = 24\ min$ $\quad \Delta t = 24 + 24\ min = 48\ min = 0.8\ hr$

a) $V_{av} = \frac{\Delta d}{\Delta t} = \frac{2.4}{0.8} = 3\ km/h$

b) Farthest pos $= 1.2\ km\ [E]$ $\quad \Delta t = 24\ min = 0.4\ hr$

$\vec{V}_{av} = \frac{\Delta \vec{d}}{\Delta \vec{t}} = \frac{1.2\ km\ [E]}{0.4} = 3\ km/h\ [E]$

c) $\vec{V}_{av} = \frac{\Delta \vec{d}}{\Delta \vec{t}} = \quad \Delta \vec{d} = 1.2\ km\ [E] + (-1.2\ km\ [E])$

$\Delta \vec{d} = 0$

$\vec{V}_{av} = \frac{0}{0.8\ hr} = 0\ km/h.$

d) Yes because the total displacement $= 0\ km$.

2. A bus driver begins a descent down a steep hill and suddenly sees a deer about to cross the road. He applies the brakes. During the bus driver's 0.32 s reaction time, the bus maintains a constant velocity of 27 m/s [forward]. Determine the displacement of the bus during the time the driver takes to react. T/I [ans: 8.6 m [forward]]

$27 \times 0.32 = 8.64$

$\vec{V} \times \Delta t = 8.6$

3. Drivers at the Daytona 500 Speedway in Florida must complete 200 laps of a track that is 4.02 km long. Calculate the average speed, in kilometres per hour, of a driver who completes 200 laps in 6.69 h. T/I [ans: 1.20×10^2 km/h]

$\Delta d = 200 \times 4.02 = 804$

$V_{av} = \frac{804}{6.69 h} = 120.2$ km/h

4. A student in a mall walks 140 m [E] in 55 s to go to his favourite store. The store is not open yet, so he walks 45 m [W] in 21 s to go to another store. Calculate his
   (a) average speed [ans: 2.4 m/s]
   (b) average velocity T/I [ans: 1.3 m/s [E]]

$\xrightarrow{140 m [E]}$ $t = 55 s$

$\xleftarrow{45 m [W]}$ $t = 21$

$\Delta d = 185$

$\Delta \vec{d} = 95$ km [E]

$\Delta t = 76 s$

$V_{av} = \frac{\Delta d}{\Delta t} = \frac{185}{76} = 2.4$ m/s

$\vec{V}_{av} = \frac{\Delta \vec{d}}{\Delta t} = \frac{95}{76} = 1.25$ m/s [E]

5. A delivery truck heads directly south for 62 km, stopping for an insignificant amount of time, and then travels 78 km directly north. The average speed for the entire trip is 55 km/h. K/U T/I C
   (a) Determine the average velocity for the entire trip in kilometres per hour. [ans: 6.3 km/h [N]]
   (b) Why is the average velocity of the truck so much smaller than the average speed?

$\Delta d = 140$ Km

$\Delta \vec{d} = 16$ km [N]

$V_{av} = 55$ km/h

$\Delta t = \frac{\Delta d}{V_{av}} = \frac{140}{55} = 2.55$

$\vec{V}_{av} = \frac{16}{2.5} = 6.27$ km/h [N]

b) $\Delta \vec{d}$ is much less than Δd.

Practice

1. Examine the position–time graphs in **Figure 7**. K/U A
   (a) In which graph(s) does the velocity increase with time? [ans: (c)]
   (b) In which graph(s) does the velocity decrease with time? [ans: (b)]

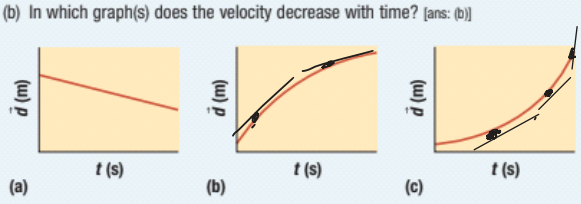


Figure 7

2. Analyze the graphs in **Figure 8**. Create a corresponding velocity–time graph for each graph. T/I C

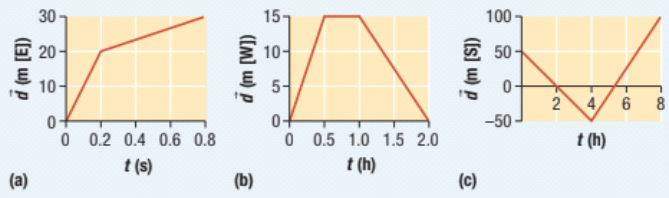


Figure 8

1a) C

1a) c

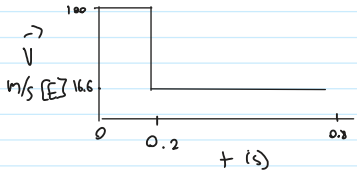
b) b

2)

2a) $t_0 = 0$

$t_1 = 0.2$ Slope $= \frac{20}{0.2} = 100$

$t_2 = 0.8$ Slope $= \frac{10}{0.6} = 16.6$

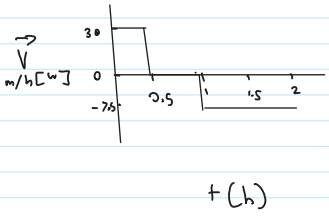


b) $t_0 = 0$

$t_1 = 0.5$ Slope $= \frac{15}{0.5} = 30$

$t_2 = 1.0$ Slope $= \frac{0}{0.5} = 0$

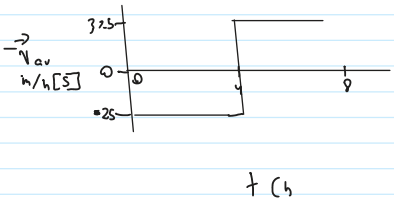
$t_3 = 2$ Slope $= \frac{-15}{2} = -7.5$



c) $t_0 = 0$

$t_1 = 4$ Slope $= \frac{-100}{4} = -25$

$t_2 = 8$ Slope $\frac{150}{4} = 37.5$



1. **Figure 10** shows a graph of the motion of a car along a straight road. K/U T/I C

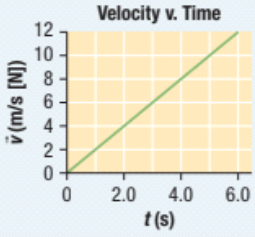


Figure 10

(a) How can you tell from the graph that the car has a constant acceleration?

(b) Describe the motion of the car.

(c) Determine the acceleration of the car. [ans: 2.0 m/s² [N]]

1a) The velocity is increasing at a constant rate.

b) ?

c) Acceleration = Slope of line

$= \frac{12}{6} = 2\ m/s^2\ [N]$

2)

2. Examine the velocity–time graph in **Figure 11**. K/U C A

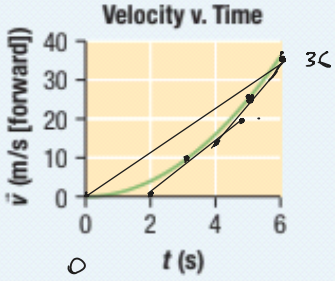


Figure 11

(a) Determine the average acceleration for the entire trip. [ans: 6 m/s²]

(b) Determine the instantaneous acceleration at 3 s and at 5 s. [ans: 6 m/s²; 10 m/s²]

(c) Draw a reasonable acceleration–time graph of the motion.

2a) $\vec{a}_{avg}$ = Slope of secant

$= \frac{36}{6} = 6\ m/s^2$ [forward]

b) Inst acceleration $= \frac{\Delta v}{\Delta t} =$

↳ @ 3s $= \frac{20-0}{4.5-2} = \frac{20}{2.5} = 8\ m/s^2$

↳ @ 5 $= \frac{35-15}{6-4} = \frac{20}{2} = 10\ m/s^2$

C) now tgm

Questions

Questions

1. A cardinal flies east for 2.9 s in a horizontal plane for a distance of 22 m from a fence post to a bush. It then flies north another 11 m to a bird feeder for 1.5 s. T/I
   (a) Calculate the total distance travelled.
   (b) Calculate the cardinal's average speed.
   (c) Calculate the cardinal's average velocity.
2. At a sled race practice field in North Bay, Ontario, a dogsled team covers a single-lap distance of 2.90 km at an average speed of 15.0 km/h. T/I
   (a) Calculate the average speed in metres per second.
   (b) Calculate the time, in seconds, needed to complete the lap.
3. A skater travels straight across a circular pond with a diameter of 16 m. It takes her 2.1 s. T/I
   (a) Determine the skater's average speed.
   (b) How long would it take the skater to skate around the edge of the pond at the same average speed?
4. An airplane flies 450 km at a compass heading of 85° for 45 min. T/I
   (a) Calculate the airplane's average speed.
   (b) Calculate the airplane's average velocity.
5. A model rocket accelerates from rest to 96 km/h [W] in 4.1 s. Determine the average acceleration of the rocket. T/I
6. A batter hits a baseball in a batting-practice cage. The ball undergoes an average acceleration of 1.37×10^3 m/s^2 [W] in 3.12×10^{-2} s before it hits the cage wall. Calculate the velocity of the baseball when it hits the wall. T/I A
7. A track runner begins running at the starting whistle and reaches a velocity of 9.3 m/s [forward] in 3.9 s. Calculate the runner's average acceleration. T/I
8. The position–time graph in **Figure 13** represents the motion of a race car moving along a straight road. K/U T/I C

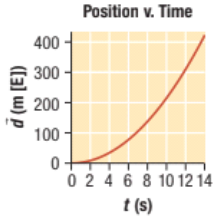


Figure 13

   (a) Determine the average velocity for the entire trip.
   (b) Determine the average velocity for the last 10 s of the motion. Why are the two average velocities different?
   (c) Determine the instantaneous velocity at 4.0 s, 8.0 s, and 12.0 s.
   (d) Sketch a qualitative velocity–time graph for the motion of the car.
9. Study the graph in **Figure 14**. K/U T/I C

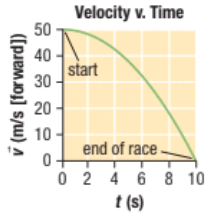


Figure 14

   (a) Determine the average acceleration for the entire trip.
   (b) Determine the instantaneous acceleration at 3.0 s, 6.0 s, and 9.0 s.
   (c) Sketch a qualitative acceleration–time graph of the motion.

1)

1. A cardinal flies east for 2.9 s in a horizontal plane for a distance of 22 m from a fence post to a bush. It then flies north another 11 m to a bird feeder for 1.5 s. T/I
   (a) Calculate the total distance travelled.
   (b) Calculate the cardinal's average speed.
   (c) Calculate the cardinal's average velocity.

1st

$t_1 = 2.9$ s

$\Delta d = 22$ m [E]

2nd

$t_2 = 1.5$ s

$\Delta \vec{d} = 11$ m [N]

$\Delta t = 2.9 + 1.5 = 4.4$

a) $\Delta d = d_1 + d_2$

$= 22 + 11$

$= 33$ m

b) $V_{avg} = \frac{\Delta d}{\Delta t}$

$= \frac{33}{4.4} = 7.5$ m/s

c) $\vec{V}_{av} = ?$

$\frac{\vec{\Delta d}}{\Delta t}$

$\vec{\Delta d} = \sqrt{11^2 + 22^2}$

$\vec{\Delta d} = 24.6$ m

$\frac{24.6 \text{ m}}{4.4 \text{ s}} = 5.6$ m/s

$\theta = \tan^{-1}\left(\frac{\Delta d_x}{\Delta d_y}\right) = \tan^{-1}\left(\frac{22}{11}\right) = 63.4$

$\vec{V}_{avg} = 5.6$ m/s [N 63.4 E]

2. At a sled race practice field in North Bay, Ontario, a dogsled team covers a single-lap distance of 2.90 km at an average speed of 15.0 km/h. T/I

2. At a sled race practice field in North Bay, Ontario, a dogsled team covers a single-lap distance of 2.90 km at an average speed of 15.0 km/h. T/I
   (a) Calculate the average speed in metres per second.
   (b) Calculate the time, in seconds, needed to complete the lap.

2) $\Delta d = 2.9\ km = 2900\ m$

a) $V_{avg} = 15\ km/h = 4.17\ m/s$

b) $V_{avg} = \frac{\Delta d}{\Delta t}$ $\quad \Delta t = \frac{\Delta d}{V_{avg}}$

$\Delta t = \frac{2900}{4.17} = 696\ s$

3. A skater travels straight across a circular pond with a diameter of 16 m. It takes her 2.1 s. T/I
   (a) Determine the skater's average speed.
   (b) How long would it take the skater to skate around the edge of the pond at the same average speed?

3 a) $\Delta d = 16\ m$

$\Delta t = 2.1\ s$

$V_{avg} = \frac{16}{2.1} = 7.62\ m/s$

b) $Circum = \pi d$

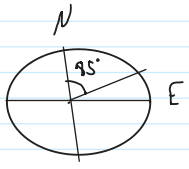
$= 3.14(16)$

$= 50.2\ m$

$\Delta t = \frac{\Delta d}{V_{avg}} = \frac{50.2}{7.62} = 6.6\ s$

4) average speed:

4. An airplane flies 450 km at a compass heading of 85° for 45 min. T/I
   (a) Calculate the airplane's average speed.
   (b) Calculate the airplane's average velocity.



$\Delta d = 450\ 000\ m$

$\Delta t = 2700\ s$

a) $V_{avg} = 166.6\ m/s$

b) $166.6\ m/s\ [N\ 85°\ E]$

$Km/h \times 0.2778 = m/s$

5) 5. A model rocket accelerates from rest to 96 km/h [W] in 4.1 s. Determine the average acceleration of the rocket. T/I

$t_1 = 0$ $\quad t_2 = 4.1$

$V_1 = 0$ $\quad V_2 = 27\ m/s\ [W]$

$\vec{a}_{avg} = \frac{27-0}{4.1-0} = 6.58\ m/s^2\ [W]$

6) 6. A batter hits a baseball in a batting-practice cage. The ball undergoes an average acceleration of 1.370×10^3 m/s² [W] in 3.12×10^{-2} s before it hits the cage wall. Calculate the velocity of the baseball when it hits the wall. T/I A

$\vec{a}_{avg} = 1370\ m/s^2\ [W]$

$\Delta t = 0.0312\ s$

$V_f = a(t) + \cancel{V_i}$

$V_f = 1370(0.0312)$

$= 42.74$ m/s [W]

7)

7. A track runner begins running at the starting whistle and reaches a velocity of 9.3 m/s [forward] in 3.9 s. Calculate the runner's average acceleration. T/I

$V_i = 0$ m/s [f]

$V_f = 9.3$ m/s [forward]

$\Delta t = 3.9$ s

$\vec{a}_{avg} = \frac{V_f}{\Delta t}$

$= \frac{9.3}{3.9}$

$\vec{a}_{avg} = 2.38$ m/s^2 [forward]